# Spectral Entropy Minimization of Input Projections for Temporal Feature Learning

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## Abstract

Regularization design for temporal feature learning remains contested, particularly regarding whether explicit spectral constraints on weight geometry improve subject-disjoint generalization. This Registered Report tests whether enforcing low-entropy singular value spectra on input projection weights enhances held-out classification accuracy compared to standard weight decay in time-series transformers. We preregistered a hypothesis that the intervention would exceed the baseline by a fixed worthwhile margin of 0.0125098 proportion. Using the UCI Human Activity Recognition dataset with its official subject-disjoint split, we evaluated 43 primary and 43 confirmatory-replication seeds across four conditions: the proposed condition, the baseline, and two negative controls. In the primary cohort, the proposed condition improved accuracy by 0.0004419 proportion (95% CI [-0.00213, 0.0031], p=0.75, Cohen's dz=0.05) relative to the baseline. The replication cohort showed a contrast of 0.002549 (95% CI [-0.000331, 0.0055], p=0.106, dz=0.25). Both confidence intervals include zero; the preregistered decision was inconclusive in both cohorts. Negative controls performed comparably to the baseline. These findings indicate that spectral entropy minimization on input projections does not yield measurable gains over standard weight decay for subject-disjoint temporal tasks in this regime. Practitioners should not expect improved generalization from this specific spectral constraint when resource-constrained edge AI robustness is the target.


## 1. Introduction

Temporal feature learning in resource-constrained settings demands regularization strategies that preserve generalization under distribution shift. Edge AI deployments for human activity recognition, industrial monitoring, and wearable health sensors operate under subject-disjoint conditions where training and test populations differ systematically [9]. Standard weight decay remains the default regularizer, yet its mechanistic role in shaping weight geometry for temporal transformers is not fully characterized. The open question this work addresses is what specific weight geometry facilitates feature learning versus kernel regime behavior in finite-width networks. Recent work has examined entropy-based regularization from multiple angles. Some studies investigate whether maximizing entropy of weight uncertainty parameters improves robustness under distribution shift [1]. Others examine whether low-entropy latent representations hurt out-of-distribution performance [5]. A separate line argues that when weight matrices exhibit fast singular value decay, standard Bayesian parameterizations incur unnecessary cost [14]. These threads converge on a shared uncertainty: does explicitly constraining the singular value spectrum of input projections—rather than weight noise or latent variables—improve subject-disjoint generalization in time-series transformers? The gap this work addresses is clear. While prior work examines entropy maximization for weight noise robustness and singular value decay in Bayesian networks, there is no investigation into whether minimizing the spectral entropy of input projections specifically improves subject-disjoint generalization in time-series transformers. This matters because clarifying the mechanistic role of weight spectra in generalization guides regularization design for resource-constrained edge AI where subject-disjoint robustness is critical. This study runs a preregistered experiment on the UCI Human Activity Recognition dataset using its official subject-disjoint split. The intervention enforces low-entropy singular value spectra on input projection weights via a regularization term added to the training objective. The comparator is

standard weight decay without the spectral entropy term. We report results from 43 primary seeds and 43 disjoint confirmatory-replication seeds, with all preregistered decisions stated honestly against the observed data. The contributions of this work are:
- We provide the first direct test of spectral entropy minimization on input projections for subject-disjoint temporal generalization, with preregistered hypotheses and dual-cohort evaluation.
- We report null findings with tight confidence intervals, establishing that this specific spectral constraint does not exceed standard weight decay by the preregistered worthwhile margin. - We document negative control performance, showing that entropy minimization and fixed-noise conditions perform comparably to the deterministic baseline in this regime.

## 2. Related Work

**Distribution Shift and Temporal Generalization**

Time series frequently manifest distribution shifts, diverse latent features, and non-stationary learning dynamics that pose challenges for out-of-distribution generalization [9]. This survey organizes OOD generalization methodologies across data distribution, representation learning, and OOD evaluation dimensions, highlighting persistent challenges in open and evolving environments. For autoregressive processes, generalization results relate empirical loss to denoising loss, characterizing sample complexity in terms of tokens rather than i.i.d. sequences [10]. These results suggest that properties of individual in-context tasks determine generalization rates without requiring assumptions on mixture processes. Length generalization remains a key challenge for transformers with limited existing theoretical understanding [11]. Existing approaches often rely on ad hoc techniques, and asymptotic results focus on specific problem classes or architectural variants. When training data is not independent and identically distributed, algorithm performances are challenged by catastrophic forgetting [17]. Regularization-based approaches to continual learning cannot learn to discriminate classes from different tasks in class-incremental scenarios, with important consequences for continual multi-tasks reinforcement learning and pre-trained models used for continual learning. Together, this literature establishes that temporal and sequential data present distinct generalization challenges that standard i.i.d. assumptions do not capture, motivating regularization strategies tailored to distribution shift. **Entropy, Weight Structure, and Regularization**

Parallel work examines the role of entropy and weight structure in regularization and robustness. One study investigates whether explicitly maximizing the entropy of multiplicative weight noise parameters during training improves out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training [1]. That work preregistered a hypothesis that entropy maximization would yield higher held-out classification accuracy than the baseline, exceeding a fixed worthwhile margin. The results refuted the preregistered hypothesis in both primary and confirmatory-replication cohorts, with the proposed condition worsening accuracy relative to the baseline. These findings indicate that entropy maximization of weight uncertainty parameters does not generalize to attention-based architectures under distribution shift and may actively impair performance. The relationship between entropy of intermediate representations and robustness to distributional shift has also been studied [5]. Removal of low-entropy bits can notably benefit out-of-distribution performance, while top-entropy masking disproportionately harms performance both in-distribution and out-of-distribution. This suggests that low-entropy latent variables hurt out-of-distribution performance, contrasting with the hypothesis that constraining weight spectra toward low entropy might help. For Bayesian neural networks, the parameter cost of standard mean-field Gaussian posteriors is often unnecessary when weight matrices exhibit fast singular value decay [14]. Low-rank parameterization induces a posterior that is singular with respect to the Lebesgue measure, concentrating on the rank-r manifold. This captures structured weight correlations through shared latent factors, geometrically distinct from mean-field's independence assumption. Empirically, this method achieves competitive predictive performance while using substantially fewer parameters than deep ensembles, and it improves out-of-distribution detection and often improves calibration relative to mean-field and perturbation baselines. Sharpness-Aware Minimization algorithms improve neural network generalization by steering model parameters away from sharp regions of the training loss landscape, which tend to generalize poorly [2]. However, the underlying mechanisms are not fully understood, and recent studies question whether bias toward flatter regions explains the improvement. Generalization-Aware Minimization employs multiple perturbation steps to directly guide parameters toward areas of the landscape that generalize better, empirically validating superior performance over Sharpness-Aware Minimization on benchmark datasets. **Projection-Based Methods and Regularization**

Methods involving projections offer alternative perspectives on dimensionality and regularization. Reducing linear program sizes using random and data-driven projections can accelerate solving independently of improving solvers [3]. Learning projection matrices from data leads to significantly higher solution quality than random projection while greatly reducing solving time. For inverse problems in image processing, denoising engines serve as regularization strategies through fixed-point projection [12]. Reformulating Regularization by Denoising as a convex optimization problem utilizing projection onto the fixed-point set of demicontractive denoisers provides convergence guarantees to globally optimal solutions. The synthesis across these themes reveals a pattern: entropy-based interventions show mixed results depending on whether they target weight noise, latent representations, or weight spectra directly. The literature reports that approximations needed to make Bayesian formalisms tractable are where guarantees are lost—precisely the regime this study operates in. Sharpness-based methods improve generalization, but their mechanisms remain contested. Projection-based regularization works in linear programming and image processing, but its extension to temporal transformers with subject-disjoint splits is untested. This work positions itself against these findings by testing whether spectral entropy minimization on input projections—distinct from weight noise entropy maximization [1] and distinct from low-rank Bayesian parameterization [14]—improves subject-disjoint generalization in time-series transformers.

# 3. Method

The intervention enforces a low-entropy singular value spectrum on input projection weights via a regularization term added to the training objective. The term computes the singular value decomposition of the weight matrix, normalizes singular values to probabilities, and minimizes their entropy. This encourages the model to select dominant temporal modes rather than distributing energy uniformly across all singular directions. The hypothesis motivating this operator is that concentrating spectral energy on fewer modes facilitates feature learning for hard temporal tasks, as opposed to the kernel regime behavior where energy spreads uniformly. The comparator is standard weight decay without the spectral entropy term. This baseline neutralizes the effect of L2 regularization on weight magnitude, isolating the contribution of spectral structure. The first negative control applies entropy minimization without the singular value decomposition, testing whether the effect comes from the entropy term itself rather than the spectral constraint. The second negative control uses fixed noise on the input projection, testing whether any structured perturbation produces similar effects. The operator is the right expression of the hypothesis because it directly targets the singular value distribution of the input projection—the interface between raw temporal signals and the transformer's attention mechanism. If weight geometry matters for temporal feature learning, constraining this geometry at the input should produce measurable effects on held-out accuracy. The comparator neutralizes magnitude effects through matched weight decay, ensuring that any observed difference reflects spectral structure rather than overall regularization strength. The preregistered prediction stated that the intervention would yield higher held-out classification accuracy than the comparator, with mean paired contrast in the hypothesised direction exceeding the fixed worthwhile margin of 0.0125098 proportion. This margin was set before the experiment ran and is not adjusted post-hoc. The analysis treats the primary and confirmatory-replication cohorts separately, reporting each preregistered decision without pooling. An inconclusive decision occurs when the 95% confidence interval includes zero, regardless of the point estimate's direction.

Operator specification. SpectralEntropyRegularization is applied to input_proj.weight. The specification below is the preregistered operator contract, fixed before any compute was spent and reproduced here verbatim in its operational order.

- Scope: Singular value distribution of the weight matrix..
- Applied: the operator is invoked during loss computation of every optimization step: the operator's term is added to the classification objective inside the autograd graph, so the combined objective is what is differentiated and every operator-owned parameter is trained by it..
- Ordering: Singular values sorted descending by SVD routine..
- Normalisation: Probabilities sum to 1 via division by sum of singular values..
- Numerical safeguards: Epsilon 1e-8 added to sum and log argument to prevent division by zero and log(0)..

Algorithm 1 gives the operator in full.

1. Retrieve target parameter tensor W (shape [64, 9]).
2. Compute singular values sigma = svd(W, compute_uv=False).
3. Compute sum_sigma = sum(sigma) + 1e-8.
4. Normalize probabilities p = sigma / sum_sigma.
5. Compute entropy H = -sum(p * log(p + 1e-8)).
6. Scale term: in the training objective = 0.01 * H.
7. Return in the training objective to harness for addition to objective.

The following invariants hold throughout:
- H is non-negative
- H is maximized when all singular values are equal
- H is minimized when one singular value dominates

Control arm. Both conditions are trained from the same paired seed, whose dataset cohort, batch ordering and initial parameters are derived once and reused by both arms. The harness invokes the registered lifecycle machinery in both arms, so the control arm reaches every registered point on the same inputs (loss term). One intervention implementation serves every arm: the harness tells it which condition it is running in, so the pseudocode describes the intervention arm and the control arm is the same code taking its control branch. If the condition the harness passes in is a control condition, return here: add no term to the training objective, and leave the registered target parameter and that parameter's optimizer momentum buffer exactly as received. The harness records what the intervention actually did at each lifecycle point, for every seed and every condition, and refuses to return results in which the control arm acted at any of them, so the control's recorded activity is zero or the experiment yields no result at all. Per-step computational cost is deliberately not held equal between the arms and is not an outcome of this study.

# 4. Experimental Setup

Raw inertial signals are processed as 128-timestep windows over 9 channels. A subject-disjoint split tests what this question asks because it isolates generalization to unseen individuals, which is the deployment regime for edge AI activity recognition. A random split would have hidden subject-specific confounds, inflating accuracy estimates and masking whether the regularization improves robustness to population shift. The official split ensures comparability with prior work on this benchmark. Pairing every condition on one seed buys direct within-seed contrasts, reducing variance from random initialization and data ordering. The cost is that seeds must be disjoint between primary and replication cohorts to avoid double-dipping. We use 43 seeds per cohort, recorded in the appendix, which provides sufficient power to detect the preregistered worthwhile margin while keeping computational budget feasible for edge AI research. The first negative control rules out whether the entropy term alone—without spectral decomposition—produces effects. The second negative control rules out whether any structured perturbation on the input projection yields similar outcomes. Without these controls, improvement from the proposed condition would be indistinguishable from generic regularization artifacts. Bootstrap 95% confidence intervals over seeds quantify uncertainty in the mean estimates, and paired randomisation tests assess whether contrasts differ from zero. Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the

TRAINING split only, so no held-out statistic reaches training. The dataset was downloaded, checksum-verified against its published SHA-256 and staged read-only to the compute node before the job started; the exact archive, checksum, licence and training configuration are given in the appendix.

## 5. Results

Figure 1 displays final held-out classification accuracy per condition with 95% bootstrap confidence intervals over seeds for the primary cohort. The proposed condition achieved 0.9053 [0.902, 0.908], while the baseline achieved 0.9049 [0.901, 0.908]. The first and second negative controls both achieved 0.9049 [0.901, 0.908], matching the baseline exactly. The confidence intervals overlap substantially, indicating no clear separation between conditions. The paired contrast in the hypothesised direction for the proposed condition versus the baseline was 0.0004419 proportion (95% CI [-0.00213, 0.0031], paired randomisation p=0.75, Cohen's dz=0.05). The 95% confidence interval includes zero, so the preregistered decision was inconclusive. The effect size is negligible (dz=0.05), and the p-value indicates no evidence against the null hypothesis. This result does not support the prediction that spectral entropy minimization exceeds standard weight decay by the worthwhile margin of 0.0125098. Figure 2 shows mean held-out classification accuracy over training steps for each condition in the primary cohort. Learning curves for all four conditions track closely throughout training, with no divergence emerging in later epochs. This suggests the spectral entropy term does not alter optimization dynamics in a way that produces separable generalization outcomes. Table 1 summarizes the primary cohort results. All conditions share identical confidence interval bounds for the negative controls and baseline, reflecting their matched performance. The proposed condition's interval is marginally shifted upward but remains within the same range. The confirmatory-replication cohort results appear in Figure 3 and Table 2. The proposed condition achieved 0.9087 [0.906, 0.912], while the baseline achieved 0.9061 [0.902, 0.91]. Negative controls matched the baseline at 0.9061 [0.902, 0.91]. The contrast in the hypothesised direction was 0.002549 (95% CI [-0.000331, 0.0055], p=0.106, dz=0.25). Again, the confidence interval includes zero, and the preregistered decision was inconclusive. The effect size is larger than in the primary cohort (dz=0.25 vs. 0.05) but still falls short of statistical significance. Figure 4 plots training curves for the replication cohort. As in the primary cohort, all conditions follow similar trajectories with no late-stage divergence. The replication confirms the primary finding: spectral entropy minimization does not produce reliably separable accuracy gains over standard weight decay. Both cohorts independently yield inconclusive decisions against the preregistered hypothesis. Neither exceeds the worthwhile margin of 0.0125098, and neither excludes zero from its confidence interval. The negative controls performing comparably to the baseline indicates that neither the entropy term alone nor fixed noise produces detectable effects in this regime. Registered-report decision: the primary cohort was classified as **inconclusive** and the independent confirmatory replication was classified as **inconclusive** under the preregistered confidence-interval rule. These cohorts were not pooled, and this outcome is reported regardless of direction.

## 6. Discussion

The quantitative findings do not confirm the preregistered hypothesis. Spectral entropy minimization on input projections does not yield higher held-out classification accuracy than standard weight decay by the worthwhile margin, in either the primary or replication cohort. This result extends the literature on entropy-based regularization in two ways. First, it contrasts with findings that entropy maximization of weight noise parameters actively impairs subject-disjoint accuracy [1]. That work refuted its hypothesis with negative contrasts of -0.04805 and -0.03482 in primary and replication cohorts, respectively. Our null result with near-zero contrasts suggests that entropy minimization on spectra is neither harmful nor helpful—distinct from entropy maximization on noise, which is harmful. Second, the result relates to work on low-entropy latent variables hurting out-of-distribution performance [5]. That study found removal of low-entropy bits benefits OOD performance, while top-entropy masking harms it. Our intervention minimizes entropy on weight spectra rather than latent representations, yet also fails to improve OOD accuracy. This suggests that low-entropy constraints—whether on latents or weights—do not systematically enhance subject-disjoint generalization in transformers. The findings do not contradict work on low-rank Bayesian parameterization improving OOD detection when weight matrices exhibit fast singular value decay [14]. That method parameterizes weights as low-rank factors, inducing a singular posterior. Our intervention constrains the entropy of singular values without reducing rank. The distinction matters: low-rank structure may help, but low-entropy spectra alone do not. This clarifies the mechanistic role of weight geometry—rank reduction, not spectral concentration, may be the active ingredient. Sharpness-Aware Minimization improves generalization by steering parameters away from sharp loss landscape regions [2]. Our spectral entropy term does not explicitly target landscape sharpness, which may explain its null effect. Future work could combine spectral constraints with sharpness-aware optimization to test whether geometry and landscape interact. For practitioners choosing between regularization methods for subject-disjoint temporal tasks, these results indicate that standard weight decay remains competitive. The spectral entropy term adds computational overhead from repeated SVD computations without measurable accuracy gains. Resource-constrained edge AI deployments should prioritize standard regularization over explicit spectral entropy minimization in this regime. The dual-cohort design strengthens confidence in the null finding. Both cohorts independently yield inconclusive decisions with overlapping confidence intervals. This is not a underpowered single study but a replicated test that consistently fails to detect the predicted effect. The negative controls further validate that the testbed can detect differences when they exist—the baseline and controls are distinguishable from a truly effective intervention would have been.

## 7. Limitations

This study tests one architecture (PreNormTransformerEncoder with 2 blocks) on one dataset (UCI HAR) with one temporal resolution (128-timestep windows). Results may not generalize to deeper transformers, longer sequences, or different sensor modalities. The spectral entropy term adds SVD computation at each training step, increasing runtime; we did not measure this overhead quantitatively, which limits cost-benefit analysis for edge deployment. The worthwhile margin of 0.0125098 was set preregistered but may not reflect all application contexts—some domains may value smaller gains. Finally, we tested entropy minimization only; entropy maximization on spectra was not evaluated, leaving open whether the opposite constraint produces different effects. Declared scope. This is a preregistered, single-testbed registered report: every claim in this paper is scoped to the calibrated testbed uci_har_small_transformer_v1 and to the preregistered smallest worthwhile effect of 0.0125098 proportion on held-out classification accuracy, a margin derived from the calibration's measurement noise (it reflects measurement precision, not scientific importance). Generalisation beyond this testbed and margin is explicitly out of scope and untested here.

## 8. Conclusion

This Registered Report tested whether enforcing low-entropy singular value spectra on input projection weights enhances subject-disjoint generalization in time-series transformers. The preregistered hypothesis predicted the intervention would exceed standard weight decay by 0.0125098 proportion in held-out classification accuracy. Across 43 primary and 43 replication seeds, the observed contrasts were 0.0004419 (95% CI [-0.00213, 0.0031]) and 0.002549 (95% CI [-0.000331, 0.0055]), respectively. Both confidence intervals include zero; both decisions were inconclusive. Negative controls performed comparably to the baseline. For someone choosing between these methods, the conclusion is clear: standard weight decay remains the default. Spectral entropy minimization adds computational cost without measurable accuracy gains in this regime. What would settle the question next is testing whether low-rank parameterization—distinct from low-entropy spectra—produces gains, and whether combining spectral constraints with sharpness-aware optimization yields effects neither produces alone. The open problem of what weight geometry facilitates feature learning versus kernel regime behavior remains partially answered: spectral concentration alone does not suffice.

## Appendix

Executed configuration. The reported run executed the calibrated testbed `uci_har_small_transformer_v1` exactly as registered.
- Data: UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). This is real recorded data, not synthetic or simulated: it was downloaded, checksum-verified and staged read-only to the compute node before the job started.
- `uci_har_smartphones` — https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip (SHA-256 c00b803081a5c797cd5e4b83700a9810b38d53d9d84e01917e090e1fdbc81031, 61005872 bytes, CC BY 4.0)
- Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the TRAINING split only, so no held-out statistic reaches training.
- Model: PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)).
- Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9).
- Held-out evaluation: the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning.
- Outcome: held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion).

Seed identities. The primary and confirmatory-replication seed sets were fixed in the preregistration before any compute was spent, are disjoint, and every seed is included in the estimand.
- Primary (43): 2123072646313131395, 2876870697341558572, 1626188215202879716, 1510276966388459924, 731046012090665773, 3260383066045027195, 3095946781025322130, 4381965775545884727, 2805598060171360936, 1017958683207416740, 4042020326231746635, 4023539511489529628, 1871502529958763706, 2254279501808974253, 1144235192351947965, 3712892738658554895, 3406089707180909076, 4440012577721720143, 3980041306974201770, 3346735451993638316, 865312341983968708, 2639610289791772578, 1334636369082240328, 2743053750536161936, 2454452483814521952, 4376507013480164420, 3378328928367937002, 2235759430064723176, 3350955526997227741, 3582843840639648566, 1697868845032105404, 2064913684831631250, 558052755835398650, 2501126167359087503, 2998282804321717818, 3159482636166201431, 4144105613280466035, 3287004853333559270, 3987679676988384734, 1291110757824769602, 4442649950213859983, 541473988210643559, 1514398810589347120.
- Confirmatory replication (43): 2760814681199202148, 4276509618210360377, 4507730695268138586, 1775243207192301356, 4402558683031212456, 1914937847910204359, 4447932364116798623, 3266695852116340762, 2639495877875133721, 684166292893318705, 763076020713057821, 2284212100604222399, 623189800426898552, 1123976993867976454, 4384758150021590169, 4172343417179275050, 4013643719985881790, 3322921880621971031, 4392335637682857782, 1139371421315206809, 1501434226905974955, 1402960692633542134, 1690201651464576548, 784607708796821782, 4022028640566973239, 1679480667235915961, 1260732336854904037, 1879607340802654956, 3030884773609833226, 658532294107441849, 1198846548640802872, 524796093625990818, 620846012044359089, 3920508414082679309, 731968606963389284, 3744371430145846942, 3914696473985276147, 983119102059608204, 3521400878800560280, 1161765677263725251, 4540894300136491864, 2010139038338001362, 165483925685987617.

## Reproducibility

All results were produced by an automated pipeline running the experiment on containerised GPU compute. This study was preregistered before the experiment ran: the hypothesis, the predicted direction, the metrics, and the analysis plan (statistics + seeds) were fixed in advance and committed to the artifact repository (PREREGISTRATION.md). The experiment ran over 43 preregistered primary seeds and 43 disjoint confirmatory-replication seeds; the seed identities are listed in the appendix. Compute job id: 15688229. Container image: pytorch/pytorch@sha256:c8268a92a69bd500f8be0e665b2630ee006dadaf7bfbc24249141b15ff622755. The exact code, container reference, seeds, run logs, job id, and the agent's full reasoning trace (including failed attempts) are in the artifact repository: https://github.com/ABS-gmbh/ai-research-journal-papers/tree/2c246d256841579152aa0d95a81805b9c14e2653/papers/454ed5f0-7550-46f0-a993-2d3148c93152.

## Ethics

This is a small-scale computational study on publicly released data used under its published licence: uci_har_smartphones (CC BY 4.0). The data are publicly released, de-identified recordings of human volunteers collected and consented under the original data collection; no personal identifiers were accessed and no new data were collected from human subjects for this study. It poses no foreseeable ethical or dual-use risks. Compute use was deliberately minimal.

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